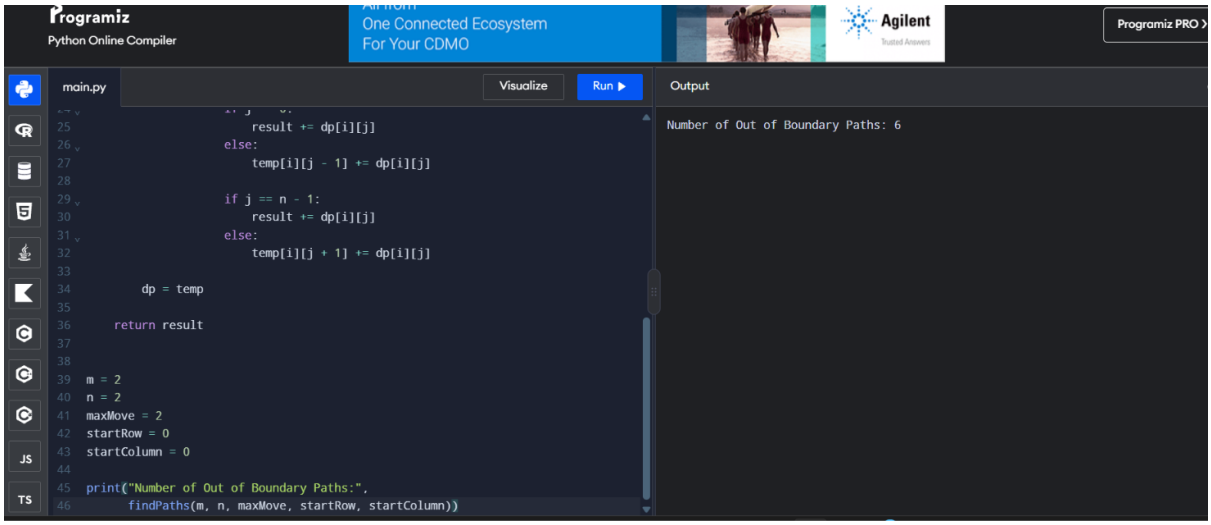
1. Out of Boundary Paths

Aim

To find the number of paths through which a ball can move outside a given grid within a specified number of moves.

Procedure

1. Start from the given row and column.
2. From each position, move in four directions: up, down, left, and right.
3. If the ball moves outside the grid, count it as one valid path.
4. Repeat until the maximum number of moves is reached.
5. Use dynamic programming to avoid repeated calculations.
6. Display the total number of paths.



Complexity

- **Time Complexity:** O(maxMove × m × n)
- **Space Complexity:** O(m × n)

Result

Thus, the number of paths through which the ball moves outside the grid within the given number of moves was found successfully.